

Basic Introduction to Number Theory for Beginners

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Preview: These notes combine beginner friendly notes of materials I studied from the textbook *A Friendly Introduction to Number Theory* by Joseph H. Silverman.

What is number theory?

Number theory is the study of the set of positive integers (also called natural numbers). In number theory we analyze patterns, prove statements, solve equations involving integers, and perform specific operations using various theorems.

Types of numbers we will encounter

- **Odd numbers:** An odd number is an integer that cannot be divided exactly by 2. When we try to divide an odd number by 2, we will always get a remainder of 1. This implies that odd numbers follow the form $n = 2k + 1$, where k is an integer and n is the odd number. It is obvious to us that we can identify any odd number by checking the last digit of any number, if the last digit is 1,3,5,7 or 9, then the number is an odd number. Some fascinating facts about odd numbers is that there are infinitely as many odd numbers, the square of an odd numbers is also an odd number and even every prime number except 2 is an odd number.

Some examples of odd numbers include: 1, 3, 5, 7, 9, ...

For $k=0$, $n=2(0)+1=1$

For $k=1$, $n=2(1)+1= 3$

- **Even numbers:** An even number is an integer that can be divided by 2. When we try to divide an even number by 2, we will always get a remainder of 0. This implies that even numbers follow the form $n = 2k$, where k is an integer and n is an even number. Likewise odd numbers, we can identify any even number by checking the last digit of any number, if the last digit is 0,2,4,6 or 8, then the number is an even number. Interestingly, even numbers contribute very less to prime numbers as there is only one even

prime number.

Some examples of even numbers include: 2, 4, 6, 8, 10, ...

For $k=0$, $n=2(0)=0$

For $k=1$, $n=2(1)=2$

- **Square numbers:** A square number is a number that results from multiplying an integer by itself. They are usually expressed in the form n^2 . An interesting fact about square numbers is that they are the sum of consecutive odd numbers or triangular numbers. They also have an odd number of factors and the last digits of every square number always end with either 0, 1, 4, 5, 6, or 9 which therefore helps us to identify them easily.

Some examples of square numbers include: 1, 4, 9, 16, 25, ...

$n=2^2=4$

$n=7^2=49$

- **Cube numbers:** A square number is a number that results from multiplying an integer by itself twice. They are usually expressed in the form n^3 . Cube numbers can usually end with any digit. Interestingly, every cube number is a centered hexagonal number. A centered hexagonal number represents a hexagon with a point in the center and additional layers of points around it.

Some examples of cube numbers include: 1, 8, 27, 64, 125, ...

$n=2^3=8$

$n=5^3=125$

- **Prime numbers:** Prime numbers are numbers that are divisible by 1 and the number itself. Prime numbers generally do not follow any specific form but some of them can be expressed in the form $n=6n\pm 1$. The Fundamental Theorem of Arithmetic states that every integer greater than 1 is either a prime itself or can be factored into a unique product of primes. These primes are called its prime factors. An interesting fact as previously mentioned, there is only one prime even number which is 2. Moreover, there are infinitely as many prime numbers. There are many theorems and problems involving prime numbers of which some are The Prime Number Theorem, The Search for Patterns: The Riemann Hypothesis, The Twin Prime Conjecture and Mersenne Primes.

Some examples of prime numbers include: 2, 3, 5, 7, 11, ... (*Note: primes are very important in number theory.*)

- **Composite numbers:** Composite numbers are integers with more than two positive divisors. In other words, it has more than two distinct positive divisors greater than 1. Composite numbers do not have any specific form but it is an very important type of number which we require in the

Fundamental Theorem of Arithmetic. A fascinating fact about composite numbers is that every composite number has a proper factor less than or equal to its square root.

Some examples of composite numbers include: 4, 6, 8, 9, 10, ...

- **Numbers congruent to 1 modulo 4:** Numbers congruent to 1 modulo 4 are integers that leave a remainder of 1 when divided by 4. They can be expressed in the form $n=4k + 1$ or $n=1+(k-1) \cdot 4$, where k is any integer. These numbers form an arithmetic progression with a common difference of 4. It is a quite important form of number we need for Fermat's Theorem where it is stated that an odd prime number p can be expressed as the sum of two squares if and only if $p \equiv 1 \pmod{4}$.

Some examples of numbers congruent to 1 modulo 4 include: 1, 5, 9, 13, 17, ...

$$n=4(0)+1=1$$

$$n=4(1)+1=5$$

- **Numbers congruent to 3 modulo 4:** Numbers congruent to 3 modulo 4 are integers that leave a remainder of 3 when divided by 4. They can be expressed in the form $n=4k + 3$ or $n=1+(k-1) \cdot 4$, where k is any integer. These numbers also form an arithmetic progression with a common difference of 4. This is the most fundamental distinction from numbers $1 \pmod{4}$. No prime number of the form $n=4k + 3$ can be expressed as a sum of two squares. This follows from Fermat's theorem on sums of two squares.

Some examples of numbers congruent to 3 modulo 4 include: 3, 7, 11, 15, 19, ...

$$n=4(0)+3=3$$

$$n=4(1)+3=7$$

- **Triangular numbers:** Triangular numbers are a sequence of numbers that represent the total number of objects that can form an equilateral triangle. They get their name from the triangular arrangements they form. They can be expressed in the form $T_n = \frac{n(n+1)}{2}$. Interestingly, the n th triangular number equals the sum of the first n natural numbers. Moreover, every perfect number is a triangular number. Also, the sum of two consecutive triangular numbers is a square number which is expressed in the form $T_{n-1} + T_n = n^2$.

Some examples of triangular numbers include: 1, 3, 6, 10, 15, ...

$$T_1 = \frac{1(1+1)}{2} = 1$$

$$T_3 = \frac{3(3+1)}{2} = 6$$

- **Perfect numbers:** A perfect number is a positive integer that is equal to the sum of its proper positive divisors (excluding the number itself). This

is one of the oldest and most intriguing concepts in number theory. They can be expressed in the form $n=2^{k-1}(2^k - 1)$ where $2^k - 1$ is a Mersenne prime, and k is an integer. They are very limited in the field of Number Theory. A fact about them is that they end with either 6 or 8. Moreover, as discussed before, the even perfect numbers are all triangular numbers.

Some examples of perfect numbers include: 6, 28, 496, 8128, 33550336, ...

$$n=2^{2-1}(2^2 - 1)=6$$

$$n=2^{3-1}(2^3 - 1)=28$$

- **Fibonacci numbers:** The Fibonacci sequence is a series of numbers where each number is the sum of the two preceding ones, usually starting with 0 and 1. They are usually expressed in the form , $T_n = T_{n-1} + T_{n-2}$. This number type has a deeper relation with the Golden Ratio and is one of the rigorously studied part of number theory.

Some examples of Fibonacci numbers include: 0, 1, 1, 2, 3, 5, ...

$$n=0+1=1$$

$$n=1+1=2$$

$$n=2+1=3$$

$$n=3+2=5$$

- **Palindromic numbers(Palindrome):** Palindromic numbers (or numeric palindromes) are numbers that remain the same when their digits are reversed. The term "palindrome" comes from the Greek words "palin" (again) and "dromos" (way, direction), meaning "running back again. Many prime numbers are palindromic numbers. These type of numbers create a sense of symmetry which helps combining geometry with number theory. They are involved in many math theorems and problems like The "Palindromic Gap" Pattern, Pascal's Triangle and The Lychrel Number Phenomenon.

Some examples of palindromic numbers include: 22, 121, 454, 1221, 2332, ...

Some properties and questions related to numbers

Below are a few interesting questions and short answers you will encounter in number theory.

1. **Can the sum of two square numbers be a square?**

Yes. Example: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$, and $5^2 + 12^2 = 25 + 144 = 169 = 13^2$.

2. **Can the sum of two n th powers be an n th power?**

In general, no. This is the statement of Fermat's Last Theorem: there are no positive integer solutions to $x^n + y^n = z^n$ for integer $n > 2$. It was proved by Andrew Wiles.

3. **Are there infinitely many prime numbers?**

Yes — classical proofs (including Euclid’s proof by contradiction) show there are infinitely many primes.

4. **Can the sum of two squares be a prime number?**

Yes. For example, $1^2 + 4^2 = 17$ and $2^2 + 3^2 = 13$.

5. **Can we relate numbers and shapes?**

Yes. Triangular numbers correspond to dot patterns forming equilateral triangles; square numbers correspond to square dot patterns, etc.

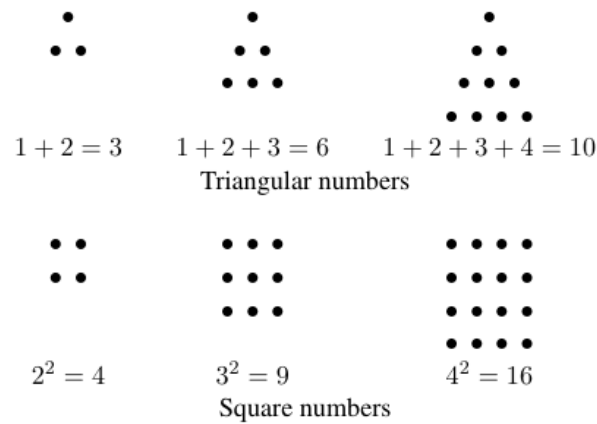


Figure 1: A graphical representation of some figurate numbers

6. **Are there infinitely many primes p such that $p + 2$ is also prime (twin primes)?**

This is the Twin Prime Conjecture: it is not yet proven whether infinitely many such primes exist. There are many known examples (3 and 5, 5 and 7, 11 and 13, etc.) and much active research on this problem.

7. **Are there prime numbers of the form $n^2 + 1$?**

Yes; some numbers of the form $n^2 + 1$ are prime (for example, $2^2 + 1 = 5$, $14^2 + 1 = 197$), though it is unknown whether there are infinitely many primes of this form.

Conclusion

Number theory is a vast field full of deep theorems, conjectures, and beautiful problems. Topics such as the Twin Prime Conjecture show how much remains to be discovered. Studying number theory strengthens mathematical reasoning and has many applications in modern fields like cryptography.